

Anomaly Detection for Smart Grids!

Tanmay Bishnoi Prateek Arora Aranan Thevendran
Carlos Mefenya Farhan Ahmed

Department of Electrical Engineering, Toronto Metropolitan University



Introduction

Energy forecasting and anomaly detection are critical components in the management of energy systems such as smart grids. Accurate energy forecasting allows utilities and organizations to predict energy demand and supply, enabling them to optimize resource allocation, reduce operational costs, and reduce carbon emissions.

Problem Statement

This project's goal is to **predict energy demand and detect anomalies in energy consumption habits** of zones and cities in Ontario using advanced time series forecasting and anomaly detection methods to allow utilities and organizations to make informed decisions in real-time.

Dataset

A combined dataset is built where the prediction target is **hourly energy demand** data for the zone/city and the predictors are **weather attributes**. We use open-source APIs provided by IESO, and Climate Canada's weather repository (ECCC), respectively, to source our data.

Table 1. Hourly dataset description

Columns	Source	Description
Target	IESO	Hourly Energy Demand of Zone/City
Predictors	ECCC	Hourly Stn Pres, Humidity, Temp

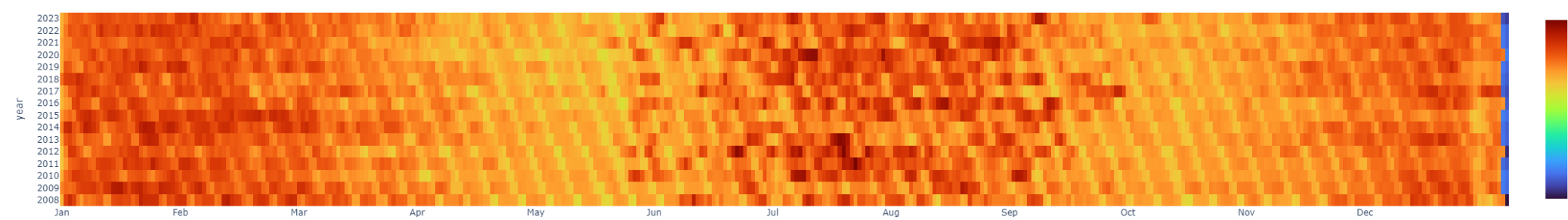


Figure 1. Seasonality of Daily energy consumption in Toronto (IESO, 2008-2023)

Approach

The solution consists of 4 components: Dataset Builder API, Forecasting Models, an Anomaly Detection Model, and an Interactive Dashboard. The user can choose the entire dataset or a subset of it to build the dataset and train the model. The results are then visualized using the dashboard.

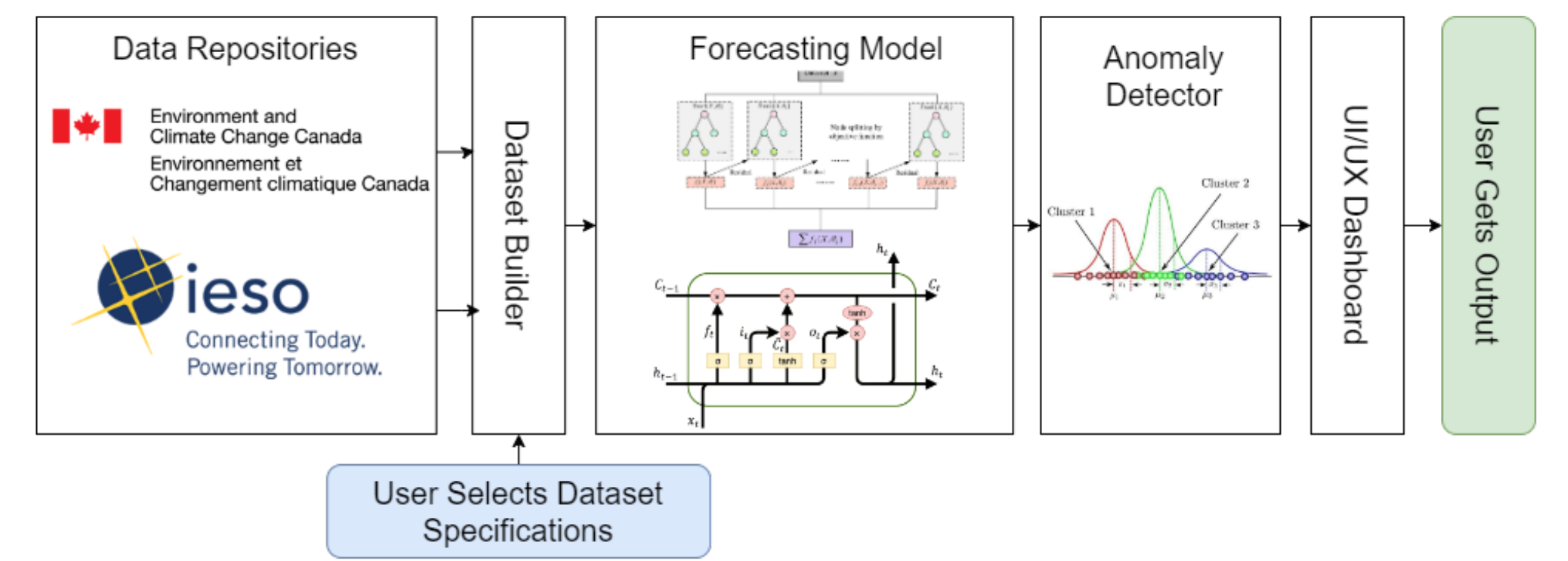


Figure 2. Full System Architecture

Forecasting Models: XGBoost

Forecasting module 1 leverages **XGBoost (Extreme Gradient Boosting)** to forecast daily energy demand based on historical consumption patterns and weather features such as temperature and humidity. XGBoost was chosen because it is a tree-based ensemble learning algorithm known for its speed and performance and interpretability.

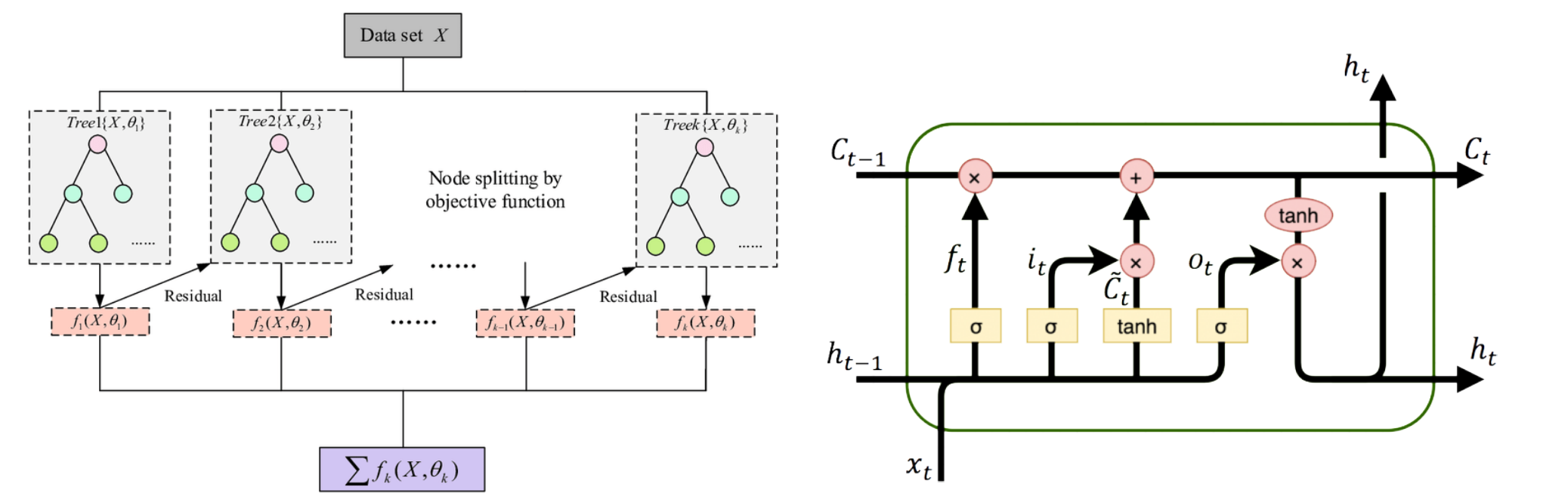


Figure 3. Model Architectures for XGBoost (left) and LSTM (right)

Forecasting Models: LSTM

Forecasting module 2 uses **LSTM (Long Short-Term Memory)**, which are Recurrent Neural Networks (RNN) designed to handle sequential data. LSTMs are highly effective for time-series forecasting as they address the vanishing gradient problem using memory cells and information gates.

Anomaly Detection Model

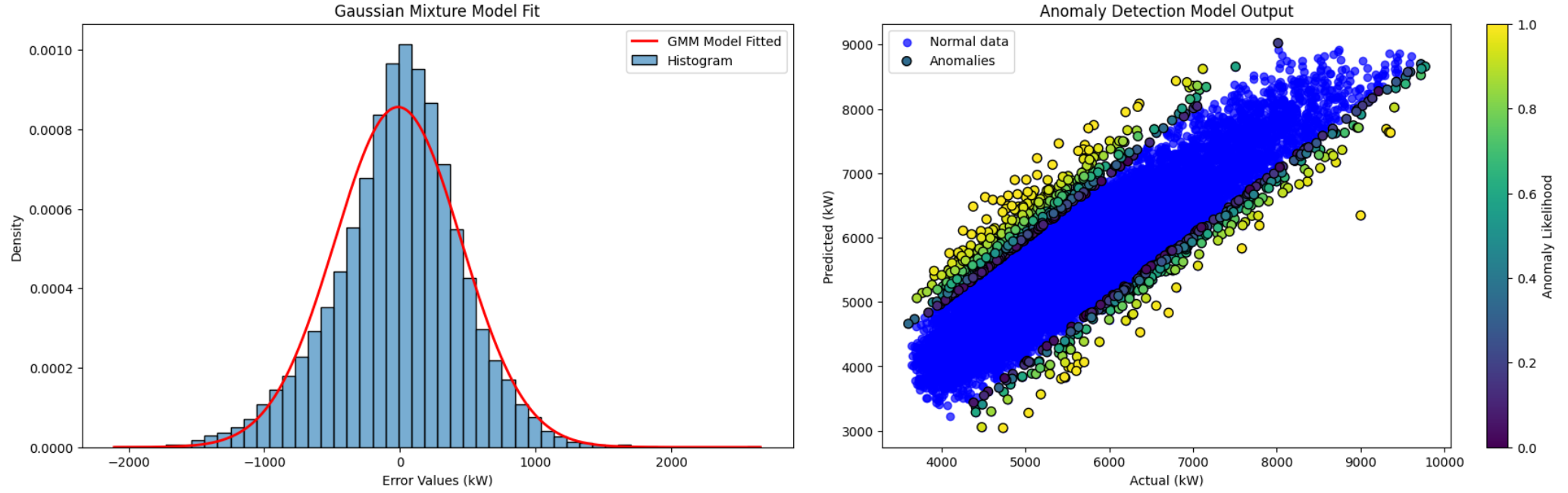


Figure 4. Gaussian Mixture Model fit (left) and Anomaly Detection Model output (right)

The **Anomaly Detection (AD)** module employs statistical and probabilistic models of the error between the predicted and actual energy demand. **Gaussian Mixture Modelling (GMM)** with **Expectation-Maximization (EM)** algorithm is used for unsupervised latent state modelling to determine the likelihood that a prediction error belongs to or falls outside of an expected distribution of errors.

Interactive Dashboard

The dashboard is a web/local application that allows the user to interact with the forecasting and anomaly detection models. The dashboard allow convenient training and inference of the models. Moreover, the user can bring their own data to train the model and test the results for anomalies.

Results

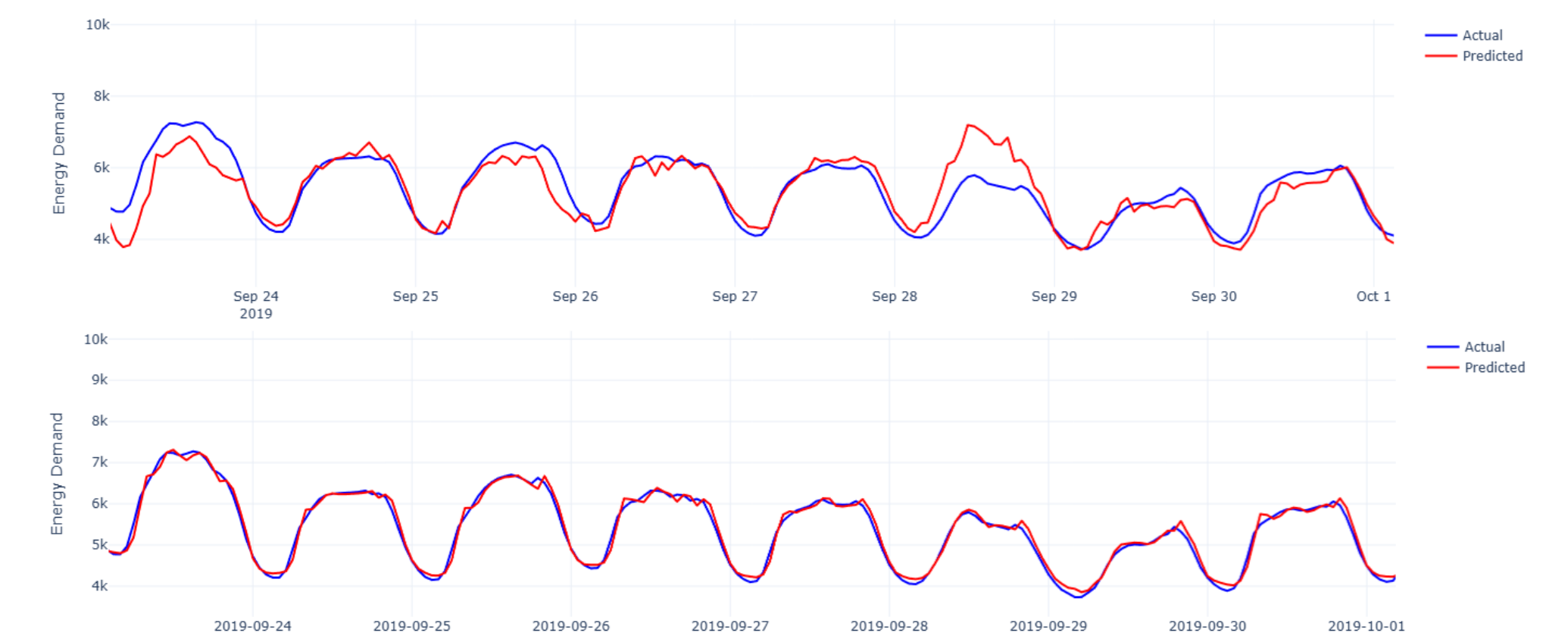


Figure 5. Hourly energy demand predictions of XGBoost (top) and LSTM (bottom) based on energy and weather data (Toronto Zonal, 2018-2021)

The models were evaluated on multiple datasets from IESO, including both Zonal (aggregated) and Forward Sortation Area (FSA) level data.

Table 2. Model performance results on multiple datasets (from 2018–2020)

Models	Zonal MAPE	Anom. %	FSA MAPE	Anom. %	Num (Datasets)
XGBoost	8.83%	4.70%	7.63%	5.0%	5
LSTM	2.44%	2.33%	4.80%	4.69%	5

Conclusion

Our research demonstrates successful implementation of energy demand forecasting and anomaly detection using advanced machine learning models. The LSTM model achieved superior performance with MAPE of **2.44%** on zonal data, while XGBoost showed robust results with **8.83%** MAPE. With our interaactive dashboards, our work can contribute to efficient energy management by making these advanced tools more accessible.

Acknowledgements

We would like to thank our supervisor, **Dr. Bala Venkatesh**, for his support throughout this project. We would also like to thank **Shima B.Z. Homayie** and **Amir R.N. Ghadikolaei** for their help and feedback.